

La Ode Muhammad Yudy Prayitno

EL12 064

Levenshtein Distance

The Levenshtein Distance or edit distance measures the similarity between two sequences/strings. The distance is measured in number of single character operations (delete, insert, replace) needed to transform one sequences into another.

Rules:

insert → remove ↓

	NULL	l	a	o	d	e
NULL	0	1	2	3	4	5
m	1	1	2	3	4	5
u	2	2	2	3	4	5
h	3	3	3	3	4	5
a	4	4	3	4	4	5
m	5	5	4	4	5	5
m	6	6	5	5	5	6
a	7	7	6	6	6	6
d	8	8	7	7	6	7

1. If $x \neq c$

Replace	Remove
Insert	$\min(\text{rep}, \text{rem}, \text{ins}) + 1$

2. If $x = c$

Just copy the diagonal element.

0	
	□

Operations for changing word "muhammad" into word "laode".

Operation	Count
Insert	
remove	III
replace	IIII
copy	I

7 operations

1. remove: << m >>
2. remove: << u >>
3. replace: << h >> to << l >>
4. remove: << m >>
5. replace: << m >> to << o >>
6. replace: << a >> to << d >>
7. replace: << d >> to << e >>

Diff: ~~muhammad~~ → laode
~~l a o d e~~ ✓ ✓